

Ibrahim Reda Wasfy

Automotive Engineer - Fresh Graduate

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📍 Obour City ,Cairo, Egypt 🛡 Military Service Exempted 🇪🇬 Egyptian 🚗 Driving License

PROFILE

Automotive engineering graduate from **Helwan University - Mataria** with a lifelong passion for motorsport that has shaped every professional choice I've made. I specialise in vehicle dynamics and steering systems, with hands-on leadership experience in two competitive teams: serving as **Head of Vehicle Dynamics** in Helwan Racing Team E-Rally, and currently **Team Leader** in HFS Racing Team. Throughout both roles, I have taken ownership of the full vehicle development cycle — spanning engineering calculations, CAD design, simulation, manufacturing, assembly, and testing — while leading subteams, coordinating workflows. Highlights include a **6th place** finish at **Shell Eco-Marathon Qatar 2025**, and **1st place** overall dynamic track at **EVER Egypt 2024**. I am a detail-oriented, motivated engineer eager to bring that competition-proven engineering mindset to a full-time role in automotive development.

EDUCATION

Bachelor Of Automotive Engineering,

Capital (Helwan) University - Mataria Branch 🔗

09/2021 – 06/2026

Cairo, Egypt

Coursework: 🔗 Vehicle Technology, Vehicle Performance, Vehicle Dynamics, Automotive Fault Diagnosis, Automotive Maintenance and Repair, Stress Analysis, Material Science, Mechanical Design, Computer Aided Design, Automotive Chassis Design, Electronic Engineering, Mechatronics, Automotive Electrical and Electronic Systems, Automatic Control, Automotive Control Systems Design.

Secondary School, *El Safwa Language School* 🔗

09/2014 – 06/2021

Obour, Cairo, Egypt

EXPERIENCE

Helwan Formula Student (HFS) Racing Team 🔗

Team Leader

06/2025 – Present

Cairo, EGY

- Led a multidisciplinary team across vehicle dynamics, powertrain, aerodynamics, and frame to develop a Formula Student prototype
- Managed project timeline, setting milestones to meet Formula Student competition deadlines (e.g., design report submission)
- Represented the team in external communications with sponsors
- Coordinated preparation for Formula Student UK, aligning technical work with competition judging criteria
- Managed all regulatory and event documentation for FSUK entry, including scrutineering forms, design report submissions.

Vehicle Dynamics

07/2025 – 07/2026

Cairo, EGY

- Directed vehicle dynamics development, including steering geometry design, Ackermann optimization, and handling performance analysis

- Mentored and trained junior members, strengthening team technical capability and ensuring knowledge continuity

Helwan Racing Team (E-Rally) [↗](#)

Head of Vehicle Dynamics

03/2025 – 01/2026

Cairo, EGY

- Led and managed the Vehicle Dynamics team.
- Set technical direction and project priorities
- Managed budgets, timelines, and resources.
- Followed up and supervised the work in the manufacturing phase.

Vehicle Dynamics Vice Head

09/2024 – 03/2025

Cairo, EGY

- Mechanical Design & Calculations for Urban & Prototype Cars
- Engineering BOM & DBR
- Preparation phase for new candidates

Vehicle Dynamics Member

07/2023 – 09/2024

Cairo, EGY

- Performed full steering and brake system calculations for a 256 kg urban EV, defining geometry and pedal ratios to meet EVER competition technical requirements.
- Mechanical design and FEA for steering parts.
- Forming a Design Report specifically in dynamics for our project

Fundraising and Public Relations

09/2024 – 01/2026

Cairo, EGY

- Secure financial support.
- Cultivate donor relationships.
- Presenting proposals

Revitup Drifting Competition [↗](#) , *Motorsport Organizer*

12/2022 – 09/2024

Cairo, EGY

- Drawing the track
- Make sure everything run smoothly
- Take care of the racer, car, audience and even the track
- Dealing with any Damage

INTERNSHIPS

Ezz-Elarab Automotive Group [↗](#) , *Engineer Intern*

09/2024

6th of Oct, Al Jizah, EGY

- Automotive Engineering courses in training center
- Attending meetings and trainings of new cars for engineers

MCV - Manufacturing Commercial Vehicles [↗](#) , *Workshop Engineer Intern*

08/2024

El Obour, Al Qalyubiyah,
Egy

- Buses, Trucks and Vans Department
- Engine Overhaul Department

ESOTICA Service Centre [↗](#) , *Workshop Engineer Intern*

07/2024

Industrial zone, New
Cairo, EGY

- Detecting electrical faults using wiring diagrams
- Service for Exotic Cars (Mclaren - Lamborghini - Porsche - Etc...)

MTI Jaguar Land Rover Egypt [↗](#) , *Workshop Engineer Intern*

06/2024

El Obour, Al Qalyubiyah,
Egy

- Completed Pre-Delivery Inspection (PDI) procedures and test plan documentation on Range Rover and Defender models

Abou Ghali Automotive [🔗](#) , *Workshop Engineer Intern*

09/2023

- Performed diesel engine services; supported workshop manager in daily job-card scheduling

El Obour, Al Qalyubiyah,
Egy

Elsaba Auto-Service Nissan [🔗](#) , *Workshop Engineer Intern*

07/2023

- Carried out scheduled maintenance services and systematic visual inspections across 10+ vehicles daily following Nissan service protocols

katameya, Cairo, Egypt

PROJECTS

Formula Student One-Seater Racing Car [🔗](#) , 2026

07/2025 – 07/2026

- Owned the complete steering system development, from concept to final design, including geometry definition, analysis, and hardware implementation
- Designed steering geometry based on Ackermann principles and tire slip characteristics, optimizing for handling performance and cornering response
- Developed the full rack and pinion steering system, including ratio selection, rack travel, and steering effort considerations
- Designed and validated a complete steering box assembly, ensuring integration with chassis and suspension constraints
- Conducted multiple design iterations to balance steering effort, responsiveness, and packaging limitations
- Integrated steering system with overall vehicle dynamics, aligning with suspension kinematics and performance targets
- Contributed to technical documentation and design justification aligned with FSUK engineering judging criteria
- Tools Used: Microsoft Excel - Solidworks - Matlab - Optimum Kinematics

Gasoline Urban Car, 2025

07/2024 – 10/2025

- Built a Gasoline Urban car and an Electric Prototype in Parallel from scratch passing by: (Calculation-Design-Manufacturing-Testing) , finishing 6th in Urban Gasoline category at Shell Eco-Marathon Qatar 2025
- Tools Used: Microsoft Excel - Solidworks

Electric Prototype Car, 2025

07/2024 – 10/2025

- Built an Electric Prototype car in Parallel with a Gasoline Urban car from scratch passing by: (Calculation-Design-Manufacturing-Testing) to participate in Shell - Eco marathon competition
- Tools Used: Microsoft Excel - Solidworks

Electric Urban Car, 2024

07/2023 – 07/2024

- Built an Electric Urban car from scratch passing by different stages mainly: (Calculation - Design - Manufacturing - Assembly - Testing), winning 1st Place Overall Dynamic Track at EVER Egypt 2024
- Tools Used: Microsoft Excel - SolidWorks

ACHIEVEMENTS

Shell - Eco marathon, Doha, Qatar

01/12/2025

- Passing Technical Inspection of Urban and prototype cars for two consecutive years
- 6th Place in Urban Gasoline Category

- 32nd Place in Electric Prototype Category

EVER Egypt, *North Coast, Egypt*

01/06/2024

- 1st Place Overall Dynamic Track
- 1st Place Design
- 1st Place Cost

ORGANISATIONS

Automobile and Touring Club Egypt (ATCE), *Time Keeper*

04/2026 – Present

Cairo, Egypt

- Operated official FIA timing systems to record and validate lap times across competitive categories
- Coordinated with race control to ensure accurate results and regulatory compliance throughout event sessions

COURSES AND CERTIFICATIONS

SolidWorks • CSWP	Matlab • Fundametals	HP Academy • Motorsport Fabrication Fundamentals • Wiring Fundametals	FIA
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TECHNICAL SKILLS

• Solidworks	• Matlab	• Simulink	• Optimum K
• AutoCAD	• Excel sheets	• Mechanical Design	• FEA Simulation
• IPG Car Maker	• Technical Report		

SOFT SKILLS

• Team leadership	• Team Work	• Multi-tasking	• Negotiation
• Project Management			

LANGUAGES

Arabic (Native)	English (Full professional proficiency)
French (Beginner)	German (Beginner)